

Pratik Kumar Biswal

Full Stack Developer | React.js | Next.js | Javascript | Node.js | Express.js | Python | FastAPI | AWS

 pkbiswal385@gmail.com |  +91 9556190293

 Portfolio: <https://pratikbiswal.vercel.app/>

 LinkedIn: <https://www.linkedin.com/in/pratikkumarbiswal/>

SUMMARY

Full Stack Software Developer with 5 years of experience building scalable web applications using React.js, Next.js, JavaScript, TypeScript, Node.js, Express.js, Python, FastAPI, PostgreSQL, MongoDB, AWS, and Docker. Skilled in developing responsive, accessible (WCAG-compliant) web interfaces, secure REST APIs, microservices, database optimization, caching, cloud deployments, and leveraging AI-assisted development tools such as GitHub Copilot to enhance productivity and code quality.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), Python, TypeScript

Frontend Technologies: React.js, Next.js, Redux Toolkit, Context API, HTML5, CSS3, Tailwind CSS, Bootstrap, Responsive Design, Server-Side Rendering (SSR), SEO Optimization, Accessibility (a11y)

Backend Technologies: Node.js, Express.js, FastAPI, RESTful APIs, API Integration, JSON Processing

Databases & Caching: PostgreSQL, MongoDB, Mongoose, Redis

Messaging & Event Streaming: Apache Kafka, Event-Driven Architecture, Message Queues, Socket.IO

Monitoring & Observability: Grafana

Authentication & Security: JWT Authentication, OAuth 2.0, Role-Based Access Control (RBAC), Bcrypt Password Hashing, Secure API Development

DevOps & Cloud: Docker, AWS, GitHub Actions, CI/CD Pipelines, Vercel

Testing & Debugging: Jest, React Testing Library, Unit Testing, Debugging, Error Handling

Artificial Intelligence & LLMs: OpenAI API, Gemini API, Prompt Engineering, GitHub Copilot

Development Practices: Agile Methodologies, Scrum, Jira, Code Reviews, Performance Optimization

Tools & IDEs: Git, GitHub, Visual Studio Code (VS Code), Postman, Jira

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Asynchronous Programming, Database Optimization, System Design Fundamentals, Microservices Architecture, Caching Strategies, Load Balancing

EXPERIENCE

Full Stack Developer | **Regal Infoways Pvt. Ltd.** | Bhubaneswar

01st June 2021 – 31st May 2026

Roles & Responsibilities

- Designed and developed scalable full stack web applications using React.js, Next.js, JavaScript, TypeScript, Node.js, Express.js, Python, and FastAPI.
- Developed responsive, role-based dashboards and reusable UI components using React.js, Next.js, HTML5, CSS3, Tailwind CSS, and Bootstrap.
- Built and maintained RESTful APIs for authentication, user management, reporting, booking workflows, billing, and real-time features using Node.js, Express.js, Python, and FastAPI.
- Implemented secure authentication and authorization mechanisms using JWT Authentication, OAuth 2.0, RBAC, and Bcrypt password encryption.

- Optimized backend performance through PostgreSQL query tuning, indexing, MongoDB schema optimization, and Redis caching strategies.
- Integrated Socket.IO for real-time notifications, live updates, and user communication features.
- Developed scalable and accessible WCAG-compliant user interfaces to enhance usability and user experience.
- Containerized applications using Docker and managed cloud deployments on AWS.
- Automated build, deployment, and release workflows using GitHub Actions and CI/CD pipelines.
- Collaborated with cross-functional teams in Agile/Scrum environments, participating in sprint planning, code reviews, debugging, and production support activities.
- Worked on performance optimization, API integration, error handling, logging, and application security best practices.
- Utilized Git, GitHub, Jira, and AI-assisted development tools such as GitHub Copilot to improve development productivity and code quality.

PROJECTS

Hospital Management System | [Regal Infoways Pvt. Ltd.](#) | 01st Aug 2022 - 31st May 2026

Description:

Developed a scalable Hospital Management System consisting of a web platform for Administrators, Doctors, and Receptionists. The system streamlined patient management, appointment scheduling, billing, prescriptions, notifications, reporting, and healthcare workflows.

Tech Stack: React.js, Next.js, JavaScript, TypeScript, Node.js, Express.js, Python, FastAPI, PostgreSQL, MongoDB, Redis, JWT, Socket.IO, Docker, AWS, Git, GitHub, Visual Studio Code

Frontend

- Used **React.js, Next.js, JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, Redux Toolkit, and Context API** to develop responsive role-based dashboards for Admin, Doctor, Receptionist, and Patients, managing over 10,000+ patient records.
- Developed appointment booking, billing, prescription management, patient registration, medical record viewing, and notification modules using **React.js, Next.js, JavaScript, and REST API integration**.
- Implemented **Server-Side Rendering (SSR), lazy loading, image optimization, and code splitting** using **Next.js**, improving page load performance by 25%.
- Built WCAG-compliant accessible UI components and responsive interfaces using **HTML5, CSS3, Tailwind CSS, and Bootstrap**, improving usability and reducing manual administrative effort by 30%.

Backend

- Designed and developed 50+ RESTful APIs for authentication, appointments, billing, patient management, and reporting using **Node.js, Express.js, Python, and FastAPI**.
- Implemented secure authentication and authorization using **JWT Authentication, RBAC, OAuth 2.0, and Bcrypt password encryption**.
- Optimized backend performance using **PostgreSQL query tuning, indexing strategies, MongoDB schema optimization, and Redis caching**, reducing query execution time by 40% and improving API response times by 30%.
- Integrated **Socket.IO** for real-time notifications, appointment updates, and patient-doctor communication.
- Used **MongoDB** for audit logging and activity tracking to improve monitoring and traceability.
- Automated deployments and CI/CD pipelines using **Docker, AWS, GitHub Actions, and Git**, reducing deployment time by 70%.

Achievements

- Heightened overall system performance by 35% through database optimization, caching strategies, frontend improvements, and accessibility-focused UI enhancements using Git, GitHub, and Visual Studio Code (VS Code).
 - Reduced production incidents by improving monitoring, centralized logging, deployment automation, and API testing with Postman.

- Delivered critical healthcare workflows supporting daily hospital operations while ensuring WCAG-compliant, accessible user experiences for healthcare professionals and patients, managed through Jira-based Agile processes.

Real Estate Platform | Regal Infoways Pvt. Ltd. | 1st June 2021 - 31st July 2022

Tech Stack: React.js, Next.js, JavaScript, Node.js, Express.js, PostgreSQL, MongoDB, Redis, JWT, Socket.IO, Docker, AWS, Git, GitHub, Visual Studio Code

Description:

Built a scalable real estate platform enabling users to search, buy, rent, and manage properties through advanced filtering, property listings, booking workflows, and role-based dashboards.

Key Contributions & Impact

Frontend

- Used **React.js, Next.js, JavaScript, HTML5, CSS3, Tailwind CSS, and Bootstrap** to build an SEO-optimized property listing and search platform, improving organic traffic visibility and user engagement.
- Developed advanced property search and filtering functionalities using **React.js, Context API, and REST API integration**, enabling users to discover properties based on location, price, amenities, and property type.
- Built responsive dashboards for Buyers, Sellers, Agents, and Administrators using **React.js and Next.js**, streamlining operational workflows and property management.
- Improved application performance by 30% using **Next.js SSR, lazy loading, image optimization, and code splitting** techniques.
- Implemented intuitive and WCAG-compliant accessible UI/UX using **HTML5, CSS3, Tailwind CSS, and Bootstrap**, improving booking workflows and customer engagement.

Backend

- Architected and developed scalable RESTful APIs for property listings, bookings, authentication, user management, and administrative operations using **Node.js, Express.js**.
- Implemented secure authentication and authorization using **JWT Authentication, RBAC, and Bcrypt password encryption** for Buyers, Sellers, Agents, and Admin users.
- Optimized backend performance using **PostgreSQL query optimization, indexing strategies, MongoDB schema design, and Redis caching**, reducing API response times by 20%.
- Integrated **Socket.IO** for real-time notifications and booking updates, improving communication between buyers, sellers, and agents.
- Designed scalable backend architecture capable of supporting growing property inventory, concurrent users, and high application traffic using **microservices concepts, Docker, and AWS**.
- Automated CI/CD pipelines and deployments using **Docker, AWS, GitHub Actions, and Git**, reducing deployment effort by 40%.

Achievements

- Modernized property search efficiency by 40% through refined filtering, database indexing, and accessibility-focused UI improvements using React.js, Next.js, PostgreSQL, Git, and GitHub.
- Reduced page load times by 30% by implementing SSR, lazy loading, code splitting, and frontend optimizations using Next.js, JavaScript, Tailwind CSS, and Visual Studio Code (VS Code).
- Delivered scalable property buying, renting, and management workflows through RESTful APIs, responsive UI development, API testing with Postman, and Agile project tracking using Jira while ensuring WCAG-compliant user experiences.

EDUCATION

B.Tech in Electrical Engineering

Indira Gandhi Institute of Technology | June 2016 – August 2020 | CGPA: 8.4/10

SOFT SKILLS

Communication | Problem-Solving | Teamwork | Time Management | Continuous Learning